Undergraduate Thesis Technical Report

# Integrated Systems Approach to Fleet Energy and Cost Analysis Using Vehicle Telematics

(technical research project in Systems Engineering)

# Lost in Translation: An Actor-Network Theory Analysis of NEPA's Permitting Network

(sociotechnical research project)


by

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On my honor as a University student, I have neither given nor received unauthorized aid on this assignment as defined by the Honor Guidelines for Thesis-Related Assignments.

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**Abstract**

Fleet electrification represents an important strategy for reducing greenhouse gas emissions while achieving long-term cost savings. Large fleets often include vehicles with diverse usage patterns, which makes uniform electrification decisions inefficient and financially risky. Decision makers therefore require analytical tools that account for these variations to evaluate economic and environmental outcomes when planning fleet transitions. This work developed a data-driven decision-support framework to evaluate electrification strategies for UVA Facilities Management (FM), which oversees approximately 300 vehicles. The framework identified which vehicles were suitable candidates for replacement with plug-in hybrid vehicles or battery electric vehicles and quantified trade-offs between cost and emissions. The methodology combined vehicle-level telematics data with a supporting machine learning based modeling approach that learned patterns in vehicle usage and energy consumption from historical telematics data for future fuel and energy estimates. Sensitivity analysis was used to evaluate uncertainty in key parameters such as fuel prices, electricity rates, charging availability, and vehicle utilization. Results indicated that a subset of high-emitting vehicles, often characterized by repeated excessive idling, contributed disproportionately to overall fleet emissions but were not always suitable candidates for electrification based on their operational profiles. Case studies of real FM replacement decisions demonstrated that while electric and hybrid alternatives offer meaningful emissions reductions of 69–82%, their higher upfront acquisition costs mean the total cost of ownership premium of electrification varies substantially by vehicle utilization, and targeted replacement strategies are more cost-effective than uniform fleet-wide electrification. These findings demonstrate that the framework provides FM with a structured, data-grounded tool to evaluate when electrification is economically justified and when operational emissions reduction should take priority.

**Keywords:** Fleet electrification, machine learning, telematics data, decision support systems, total cost of ownership, data-driven decision making